241071075
Trisha Wakode

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║         UC-01: Enroll in Course — Black Box Test Suite
         Techniques : Equivalence Class Partitioning (ECP)
                      Boundary Value Analysis (BVA)
         Focus      : Course ID validity  |  Student Eligibility
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  PART A  —  ECP: Course ID Equivalence Classes
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========================================================================
  Test  1 | TC-ECP01     | Technique : ECP — Valid partition
  Scenario  : Valid alphanumeric course ID that exists in the system
  ----------------------------------------------------------------------
  Inputs    :
              student_id               = S001
              logged_in                = True
              course_id                = C101
              course_available         = True
              seats                    = 30
  ----------------------------------------------------------------------
  Expected  : SUCCESS
  Actual    : SUCCESS: Enrollment successful [ID=E1] — Trisha Wakode enrolled in Software Engineering
  Result    : >>> PASS <<<
========================================================================

========================================================================
  Test  2 | TC-ECP02     | Technique : ECP — Valid partition
  Scenario  : Longer valid alphanumeric course ID (multi-char prefix, exists)
  ----------------------------------------------------------------------
  Inputs    :
              student_id               = S001
              logged_in                = True
              course_id                = CS202
              course_available         = True
              seats                    = 15
  ----------------------------------------------------------------------
  Expected  : SUCCESS
  Actual    : SUCCESS: Enrollment successful [ID=E1] — Trisha Wakode enrolled in Computer Science
  Result    : >>> PASS <<<
========================================================================

========================================================================
  Test  3 | TC-ECP03     | Technique : ECP — Invalid partition
  Scenario  : Empty string passed as course ID
```

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╔═══ BLACK BOX FINAL SUMMARY TABLE ═══╗

#    TC ID         Technique                          Scenario                                          Expected                        R
esult
──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────
──────────

  PART A  —  ECP: Course ID
──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────
──────────
1    TC-ECP01      ECP — Valid partition              Valid alphanumeric course ID that exists in the syst  SUCCESS                         P
ASS
2    TC-ECP02      ECP — Valid partition              Longer valid alphanumeric course ID (multi-char pref  SUCCESS                         P
ASS
3    TC-ECP03      ECP — Invalid partition            Empty string passed as course ID                      Invalid course ID               P
ASS
4    TC-ECP04      ECP — Invalid partition            Course ID contains special characters (C@#1)          Invalid course ID               P
ASS
5    TC-ECP05      ECP — Invalid partition            Numeric-only course ID (no letter prefix) — valid fo  Course not found                P
ASS
6    TC-ECP06      ECP — Invalid partition            Valid alphanumeric format but course ID does not exi  Course not found                P
ASS

  PART B  —  ECP: Student Eligibility
──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────
──────────
7    TC-ECP07      ECP — Valid partition              Student exists, logged in, not yet enrolled — all pr  SUCCESS                         P
ASS
8    TC-ECP08      ECP — Invalid partition            Student exists but is NOT logged in (precondition vi  Student must be logged in         P
ASS
9    TC-ECP09      ECP — Invalid partition            Student ID not registered in the system at all (S999  Student not found               P
ASS
10   TC-ECP10      ECP — Invalid partition            Student already enrolled in course — duplicate enrol  already enrolled                P
ASS

  PART C  —  BVA: Course ID Length & Seats
──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────
──────────
11   TC-BVA01      BVA — min (len=1)                  Course ID = 1 character (min length) — format valid,  Course not found                P
ASS
12   TC-BVA02      BVA — min+1 (len=2)                Course ID = 2 characters (min+1) — exists in system,  SUCCESS                         P
ASS
13   TC-BVA03      BVA — max (len=10)                 Course ID = 10 characters (max valid length) — exist  SUCCESS                         P
ASS
```

```
  PART C  —  BVA: Course ID Length & Seats
──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────
──────────
11   TC-BVA01      BVA — min (len=1)                  Course ID = 1 character (min length) — format valid,  Course not found                P
ASS
12   TC-BVA02      BVA — min+1 (len=2)                Course ID = 2 characters (min+1) — exists in system,  SUCCESS                         P
ASS
13   TC-BVA03      BVA — max (len=10)                 Course ID = 10 characters (max valid length) — exist  SUCCESS                         P
ASS
14   TC-BVA04      BVA — seat boundary (1)            Exactly 1 seat remaining — enrollment succeeds, seat  SUCCESS                         P
ASS
15   TC-BVA05      BVA — seat boundary (0)            0 seats available (below boundary) — alternate flow,  Course is unavailable            P
ASS
──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────
──────────

  Total : 15   |   PASS : 15   |   FAIL : 0

  All 15 test cases PASSED.


╔═══ UC-01: Enroll in Course — White Box Test Suite ═══╗
    Techniques : Statement Coverage + Branch Coverage
    Target     : enroll() control flow + helper methods
    V(G)       : 7  (7 independent basis paths)


  PART D  —  Branch Coverage: enroll() — 7 Independent Paths

======================================================================
  Test  1 | TC-WB01      | Technique : Branch — B1 False (acting student absent)
  Scenario  : Path 1: acting_student_id not in _students → LookupError
  ------------------------------------------------------------------
  Inputs    :
              acting_student_id        = S999
              registered_in_system     = False
              node_covered             = B1-F → S2 raise LookupError
  ------------------------------------------------------------------
  Expected  : Student not found
  Actual    : [Exception] LookupError: Student not found
  Result    : >>> PASS <<<
======================================================================

======================================================================
```

```
╔═══════════════════════════════════════════════════════════════╗
  UC-01: Enroll in Course — White Box Test Suite
  Techniques : Statement Coverage + Branch Coverage
  Target     : enroll() control flow + helper methods
  V(G)       : 7  (7 independent basis paths)
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  PART D  —  Branch Coverage: enroll() — 7 Independent Paths
═══════════════════════════════════════════════════════════════════

======================================================================
  Test  1 | TC-WB01      | Technique : Branch — B1 False (acting student absent)
  Scenario  : Path 1: acting_student_id not in _students → LookupError
  ----------------------------------------------------------------
  Inputs    :
              acting_student_id        = S999
              registered_in_system     = False
              node_covered             = B1-F → S2 raise LookupError
  ----------------------------------------------------------------
  Expected  : Student not found
  Actual    : [Exception] LookupError: Student not found
  Result    : >>> PASS <<<
======================================================================

======================================================================
  Test  2 | TC-WB02      | Technique : Branch — B2 False (not logged in)
  Scenario  : Path 2: acting student found but logged_in=False → PermissionError
  ----------------------------------------------------------------
  Inputs    :
              acting_student_id        = S004
              logged_in                = False
              course_id                = C101
              node_covered             = B1-T → B2-F → S3 raise PermissionError
  ----------------------------------------------------------------
  Expected  : Student must be logged in
  Actual    : [Exception] PermissionError: Student must be logged in
  Result    : >>> PASS <<<
======================================================================

======================================================================
  Test  3 | TC-WB03      | Technique : Branch — B3 False (request student absent)
  Scenario  : Path 3: acting student OK; request.student_id not found → LookupError
  ----------------------------------------------------------------
  Inputs    :
              acting_student_id        = S001
              logged_in                = True
```

```
======================================================================
  Test  3 | TC-WB03      | Technique : Branch — B3 False (request student absent)
  Scenario  : Path 3: acting student OK; request.student_id not found → LookupError
  ----------------------------------------------------------------
  Inputs    :
              acting_student_id        = S001
              logged_in                = True
              request_student_id       = S999
              registered               = False
              node_covered             = B1-T → B2-T → B3-F → S4 raise LookupError
  ----------------------------------------------------------------
  Expected  : Student not found
  Actual    : [Exception] LookupError: Student not found
  Result    : >>> PASS <<<
======================================================================

======================================================================
  Test  4 | TC-WB04      | Technique : Branch — B4 False (course absent)
  Scenario  : Path 4: both students OK; course_id 'C000' not in _courses → LookupError
  ----------------------------------------------------------------
  Inputs    :
              acting_student_id        = S001
              logged_in                = True
              course_id                = C000
              exists_in_system         = False
              node_covered             = B1-T → B2-T → B3-T → B4-F → S5 raise LookupError
  ----------------------------------------------------------------
  Expected  : Course not found
  Actual    : [Exception] LookupError: Course not found
  Result    : >>> PASS <<<
======================================================================

======================================================================
  Test  5 | TC-WB05      | Technique : Branch — B5 True (course unavailable)
  Scenario  : Path 5: all guards pass; course available=False → ERROR alternate flow
  ----------------------------------------------------------------
  Inputs    :
              acting_student_id        = S001
              logged_in                = True
              course_id                = C999
              available                = False
              seats                    = 0
              node_covered             = B1-T → B2-T → B3-T → B4-T → B5-T → S7 return ERROR
  ----------------------------------------------------------------
  Expected  : Course is unavailable
  Actual    : ERROR: Course is unavailable — redirecting to browse page
  Result    : >>> PASS <<<
```

```
  Test  6 | TC-WB06      | Technique : Branch — B6 True (already enrolled)
  Scenario  : Path 6: course available, enrollment record exists → ValueError duplicate
  ------------------------------------------------------------------
  Inputs    :
              acting_student_id        = S001
              logged_in                = True
              course_id                = C101
              pre_enrolled             = True
              node_covered             = B1-T → B2-T → B3-T → B4-T → B5-F → B6-T → S8 raise ValueError
  ------------------------------------------------------------------
  Expected  : already enrolled
  Actual    : [Exception] ValueError: Student already enrolled in this course
  Result    : >>> PASS <<<
======================================================================

======================================================================
  Test  7 | TC-WB07      | Technique : Branch — full success path (all True)
  Scenario  : Path 7: all checks pass, no duplicate → SUCCESS enrollment created
  ------------------------------------------------------------------
  Inputs    :
              acting_student_id        = S001
              logged_in                = True
              course_id                = C101
              available                = True
              seats                    = 30
              pre_enrolled             = False
              node_covered             = B1-T → B2-T → B3-T → B4-T → B5-F → B6-F → S9-S11 SUCCESS
  ------------------------------------------------------------------
  Expected  : SUCCESS
  Actual    : SUCCESS: Enrollment successful [ID=E1] — Trisha Wakode enrolled in Software Engineering
  Result    : >>> PASS <<<
======================================================================

  ------------------------------------------------------------------
  PART E  —  Branch Coverage: _check_availability() sub-paths
  ------------------------------------------------------------------

======================================================================
  Test  8 | TC-WB08      | Technique : Branch — BA False (available=False)
  Scenario  : course.available=False → _check_availability returns False → ERROR
  ------------------------------------------------------------------
  Inputs    :
              course_id                = C999
              available                = False
              seats_available          = 0
              node_covered             = BA: available=False → returns False (short-circuit)
  ------------------------------------------------------------------
```

```
  Test  9 | TC-WB09      | Technique : Branch — BB False (seats=0, available=True)
  Scenario  : available=True but seats_available=0 → _check_availability False → ERROR
  ------------------------------------------------------------------
  Inputs    :
              course_id                = C104
              available                = True
              seats_available          = 0
              node_covered             = BA: available=True → BB: seats=0 → returns False
  ------------------------------------------------------------------
  Expected  : Course is unavailable
  Actual    : ERROR: Course is unavailable — redirecting to browse page
  Result    : >>> PASS <<<
======================================================================

======================================================================
  Test 10 | TC-WB10      | Technique : Branch — BA+BB True (available & seats>0)
  Scenario  : available=True, seats=25 → _check_availability True → enrollment proceeds
  ------------------------------------------------------------------
  Inputs    :
              course_id                = C102
              available                = True
              seats_available          = 25
              node_covered             = BA: available=True → BB: seats>0 → returns True → SUCCESS
  ------------------------------------------------------------------
  Expected  : SUCCESS
  Actual    : SUCCESS: Enrollment successful [ID=E1] — Trisha Wakode enrolled in Data Structures
  Result    : >>> PASS <<<
======================================================================

  ------------------------------------------------------------------
  PART F  —  Statement Coverage: Helper Method Paths
  ------------------------------------------------------------------

======================================================================
  Test 11 | TC-WB11      | Technique : Statement — get_enrolled_courses (found)
  Scenario  : get_enrolled_courses: student S001 exists & enrolled → returns list with C101
  ------------------------------------------------------------------
  Inputs    :
              student_id               = S001
              enrolled                 = ['C101']
              node_covered             = s = _students.get(S001) → s is not None → return list
  ------------------------------------------------------------------
  Expected  : SUCCESS
  Actual    : SUCCESS
  Result    : >>> PASS <<<
======================================================================
```

```
======================================================================
  Test 12 | TC-WB12      | Technique : Statement — get_enrolled_courses (not found)
  Scenario  : get_enrolled_courses: student S999 absent → returns empty list []
  ----------------------------------------------------------------------
  Inputs    :
              student_id               = S999
              registered               = False
              node_covered             = s = _students.get(S999) → s is None → return []
  ----------------------------------------------------------------------
  Expected  : SUCCESS
  Actual    : SUCCESS
  Result    : >>> PASS <<<
======================================================================

======================================================================
  Test 13 | TC-WB13      | Technique : Statement — has_access (match found → True)
  Scenario  : has_access: S001+C101 enrollment exists → returns True
  ----------------------------------------------------------------------
  Inputs    :
              student_id               = S001
              course_id                = C101
              enrolled                 = True
              node_covered             = any(...) → match found → returns True
  ----------------------------------------------------------------------
  Expected  : SUCCESS
  Actual    : SUCCESS
  Result    : >>> PASS <<<
======================================================================

======================================================================
  Test 14 | TC-WB14      | Technique : Statement — has_access (no match → False)
  Scenario  : has_access: S001+C101 not enrolled → returns False
  ----------------------------------------------------------------------
  Inputs    :
              student_id               = S001
              course_id                = C101
              enrolled                 = False
              node_covered             = any(...) → no match → returns False
  ----------------------------------------------------------------------
  Expected  : SUCCESS
  Actual    : SUCCESS
  Result    : >>> PASS <<<
======================================================================

======================================================================
  Test 15 | TC-WB15      | Technique : Statement — _create_enrollment (ID increment)
  Scenario  :  create enrollment: sequential IDs E1, E2 assigned across two enrollments
```

```
======================================================================
  Test 15 | TC-WB15      | Technique : Statement — _create_enrollment (ID increment)
  Scenario  : _create_enrollment: sequential IDs E1, E2 assigned across two enrollments
  ----------------------------------------------------------------------
  Inputs    :
              student_id_1             = S001
              student_id_2             = S002
              course_id                = C101
              node_covered             = eid=f'E{_next_enroll_id}' → id++ → Enrollment appended
  ----------------------------------------------------------------------
  Expected  : SUCCESS
  Actual    : SUCCESS
  Result    : >>> PASS <<<
======================================================================
```

```
╔═══ WHITE BOX FINAL SUMMARY TABLE ═══╗

#    TC ID         Technique                    Scenario                                  Expected output              Actual output
                         Result
──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────
──────────────────────────────────────

  PART D  —  Branch Coverage: enroll() — 7 Paths (V(G)=7)
──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────
──────────────────────────────────────
1    TC-WB01       Branch — B1 False (acting st  Path 1: acting_student_id not in _students  Student not found             [Exception] LookupError:
Student not found        PASS
2    TC-WB02       Branch — B2 False (not logge  Path 2: acting student found but logged_in  Student must be logged in     [Exception] PermissionErr
or: Student must be log  PASS
3    TC-WB03       Branch — B3 False (request s  Path 3: acting student OK; request.student  Student not found             [Exception] LookupError:
Student not found        PASS
4    TC-WB04       Branch — B4 False (course ab  Path 4: both students OK; course_id 'C000'  Course not found              [Exception] LookupError:
Course not found         PASS
5    TC-WB05       Branch — B5 True (course una  Path 5: all guards pass; course available=  Course is unavailable         ERROR: Course is unavaila
ble — redirecting to br  PASS
6    TC-WB06       Branch — B6 True (already en  Path 6: course available, enrollment recor  already enrolled              [Exception] ValueError: S
tudent already enrolled  PASS
7    TC-WB07       Branch — full success path (  Path 7: all checks pass, no duplicate → SU  SUCCESS                       SUCCESS: Enrollment succe
ssful [ID=E1] — Trisha   PASS

  PART E  —  Branch Coverage: _check_availability()
──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────
──────────────────────────────────────
8    TC-WB08       Branch — BA False (available  course.available=False → _check_availabili  Course is unavailable         ERROR: Course is unavaila
ble — redirecting to br  PASS
9    TC-WB09       Branch — BB False (seats=0,   available=True but seats_available=0 → _ch  Course is unavailable         ERROR: Course is unavaila
ble — redirecting to br  PASS
10   TC-WB10       Branch — BA+BB True (availab  available=True, seats=25 → _check_availabi  SUCCESS                       SUCCESS: Enrollment succe
ssful [ID=E1] — Trisha   PASS

  PART F  —  Statement Coverage: Helper Methods
──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────
──────────────────────────────────────
11   TC-WB11       Statement — get_enrolled_cou  get_enrolled_courses: student S001 exists   SUCCESS                       SUCCESS
                         PASS
12   TC-WB12       Statement — get_enrolled_cou  get_enrolled_courses: student S999 absent   SUCCESS                       SUCCESS
                         PASS
13   TC-WB13       Statement — has_access (matc  has_access: S001+C101 enrollment exists →   SUCCESS                       SUCCESS
                         PASS
14   TC-WB14       Statement — has access (no m  has access: S001+C101 not enrolled → retur  SUCCESS                       SUCCESS

                                                                                                   Ctrl+K to generate command
```

```
7    TC-WB07       Branch — full success path (  Path 7: all checks pass, no duplicate → SU  SUCCESS                        SUCCESS: Enrollment succe
ssful [ID=E1] — Trisha   PASS

  PART E  —  Branch Coverage: _check_availability()
──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────
──────────────────────────────────────
8    TC-WB08       Branch — BA False (available  course.available=False → _check_availabili  Course is unavailable         ERROR: Course is unavaila
ble — redirecting to br  PASS
9    TC-WB09       Branch — BB False (seats=0,   available=True but seats_available=0 → _ch  Course is unavailable         ERROR: Course is unavaila
ble — redirecting to br  PASS
10   TC-WB10       Branch — BA+BB True (availab  available=True, seats=25 → _check_availabi  SUCCESS                        SUCCESS: Enrollment succe
ssful [ID=E1] — Trisha   PASS

  PART F  —  Statement Coverage: Helper Methods
──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────
──────────────────────────────────────
11   TC-WB11       Statement — get_enrolled_cou  get_enrolled_courses: student S001 exists   SUCCESS                        SUCCESS
                        PASS
12   TC-WB12       Statement — get_enrolled_cou  get_enrolled_courses: student S999 absent   SUCCESS                        SUCCESS
                        PASS
13   TC-WB13       Statement — has_access (matc  has_access: S001+C101 enrollment exists →   SUCCESS                        SUCCESS
                        PASS
14   TC-WB14       Statement — has_access (no m  has_access: S001+C101 not enrolled → retur  SUCCESS                        SUCCESS
                        PASS
15   TC-WB15       Statement — _create_enrollme  _create_enrollment: sequential IDs E1, E2   SUCCESS                        SUCCESS
                        PASS
──────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────────
──────────────────────────────────────

  Total : 15   |   PASS : 15   |   FAIL : 0

  All 15 test cases PASSED.

exam1@sysadmin-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/Desktop/seLab$ git add .
exam1@sysadmin-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/Desktop/seLab$ git commit -m "white box testcases"
[master 8cd34b5] white box testcases
 1 file changed, 332 insertions(+), 17 deletions(-)
exam1@sysadmin-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/Desktop/seLab$ git push
To https://github.com/trishawakode1/seLab_CourseEnrollment.git
 ! [rejected]        master -> master (fetch first)
error: failed to push some refs to 'https://github.com/trishawakode1/seLab_CourseEnrollment.git'
hint: Updates were rejected because the remote contains work that you do
hint: not have locally. This is usually caused by another repository pushing
hint: to the same ref. You may want to first integrate the remote changes
hint: (e.g., 'git pull ...') before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
exam1@sysadmin-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/Desktop/seLab$
```

Ctrl+K to generate command